



SC30 Shutter Controller

User Guide



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Chapter 1 Description

1.1 Intended Use

Thorlabs SC30 shutter controller is designed to operate Thorlabs series SHx bladed shutters and SHBx series iris shutters. These shutters are intended to control the exposure of light through a system whether it be lasers for the bladed shutters or images for the iris shutters. It is intended to operate in typical laboratory environments.

1.2 Explanation of Safety Warnings

NOTICE

Indicates information considered important, but not hazard-related, such as possible damage to the product.



Danger, Warning, or Caution



Shock Warning



The CE/UKCA markings on the product are the manufacturer's declaration that the product complies with the essential requirements of the relevant European health, safety, and environmental protection legislation.



The wheeled bin symbol on the product, the accessories or packaging indicates that this device must not be treated as unsorted municipal waste but must be collected separately.

1.3 Description

Thorlabs SC30 is a controller designed to operate all Thorlabs series of shutters including both blade and iris shutters. The controller can operate 2 devices at the same time either independently or synced to operate together. The SC30 provides a direct interface to the SHx series blade shutters and the SHBx series iris shutter, providing the drive levels required to operate the shutter with millisecond accuracy.

In addition, the SC30 also provides front panel TRIGGER's that can be configured as inputs or outputs depending on the mode and users desired configuration. The unit can be triggered from a TTL input. By default, TRIGGER1 is set up as a trigger input and TRIGGER2 is configured as an output. Some modes will force trigger configurations but generally they can be set as inputs or outputs. Output triggers can be set to show either channel, the actual drive signal, or the monitor feedback from the attached device.

An LCD display with touchscreen allows for complete control over shutter operation. The front control panel allows the user to operate the device in 4 different modes. Manual mode is a simple touch interface to open and close the shutter with direct input from the user. External mode utilizes the TRIGGER1 input to open and close the shutter based on the voltage level of the external signal. High opens the shutter while low closes it. Internal mode allows the user to set up a continuous open/close cycle based on the open and close times settings. Triggered mode allows for specific timing, number of cycles, and triggering to control the conditions of the shutter.

The unit may also be controlled and operated from a remote USB interface. All the provisioning and control can be programmed using a command line editor, or from the stand-alone PC application software supplied with the unit. The unit also comes with Lab Windows/CVI® and LabVIEW® drivers. Ethernet connections are available but not yet operational. This functionality will be included in future updates.

When used with Thorlabs shutters, the unit detects the presence of the shutter and provides an alarming/monitor feature to ensure the status of the shutter agrees with the output of the controller.

An interlock function is also provided on the SC30. It can be used with a switch to close shutters as a safety feature when a door is opened. This functionality can be enabled or disabled through the settings menu.

The SC30 includes a touchscreen LCD controller to provide simple and intuitive adjustments of all parameters. An additional encoder allows for fine control of each of the settings and is backlit with color coded LEDs to the selected parameters. Along with the front panel touchscreen adjustments, all functionality is now available through a USB2.0 connection. The device can be fully adjusted remotely and includes save/recall settings convenient when using multiple Interferometers.

This device is not designed according to the requirements of IEC 61508 and therefore is not a safety component. The product may only be used in accordance with the instructions described in this manual. Any other use will invalidate the warranty.

1.4 Technical Data

1.4.1 Specifications

General Specifications		
Output Characteristics		
Trigger Connectors		BNC (Input/Output)
Trigger Out Voltage		5 V TTL Typical
Trigger Out Current		32 mA max
Trigger In Range	5 V TTL	
	High	2.0 V Min, 5.0 V Max
	Low	0.5 V Max
Trigger In/Out Impedance ^a		100 Ω
Shutter Output Connectors	SHBx Series	4-pin mini-DIN
	SHx Series	Hirose HR10-7R-6S
Shutter Output Drive Voltage	SHBx Series (DIN Connector)	7.0 V
	SHx Series (Hirose Connector)	24.0 V
Shutter Output Short Circuit Current ^b		1.67 A
Shutter Output Drive Type		Pulse Width Modulation (PWM)
Shutter Drive PWM Frequency		22 kHz
Shutter Performance		
Maximum Exposure Rate ^c		50 Hz
Minimum Exposure Time ^d		10 ms
Typical Transfer Time		1 ms
Accuracy		0.2 ms
On/Off Times ^{d,e}		10 ms to 1000 s
Maximum Steady State Power		8 W per Output
Average Hold Voltage Range	SHBx Series	1.3 to 1.5 V PWM
	SHx Series	9.1 to 10.1 V PWM
Voltage Output	SHBx Series	7 V Pulse
	SHx Series	24 V Pulse
Output Enable		2.5 mm Interlock Connector, Touchscreen Button

a. The inline protection resistance included between the BNC connection and the circuitry adds to the external load.

b. The output is short circuit protected above the output current limit.

c. Highest Frequency at which the Controller can Perform an On/Off Sequence

d. If the on/off time is set to a value lower than the minimum exposure time of the unit, the shutter may not open or close fully.

e. The amount of time that the shutter will remain open or closed is set in the settings panel and is valid when operating in auto mode.

Timing Specifications		
Event	SH05R(/M) (Typical / Maximum)	SH1(/M) (Typical / Maximum)
TI	8.0 ms / 10 ms	10 ms / 20 ms
TO	4.3 ms / 6 ms	10 ms / 20 ms
TD/R	11.0 ms / 20 ms	20 ms / 30 ms
TC	6.5 ms / 8 ms	10 ms / 20 ms
MOP	10 ms / 10 ms	40 ms / 40 ms
MSOP	14 ms / 14 ms	40 ms / 40 ms
Timing Definitions		
Event	SH05R(/M)	SH1(/M)
TI	Time Initialize: Time from Initialization pulse, to energizing of coil, to the initial movement of the shutter.	Transfer Initialize: the time delay between the application of the energizing voltage and the initial movement of the shutter.
TO	Time Open (Rise): Time of shutter blade movement to open from 10% exposure ^a to 90% exposure.	Transfer Open: the time for the shutter to move from 20% exposure ^a to 80% exposure.
TD/R	Time Delay Release: Time delay of release of the hold voltage or pulse at the moment the shutter blade starts to close.	Transfer Dwell/Release: the delay between the removal of the energizing voltage and the initial closing movement of the shutter.
TC	Time Close (Fall): Time of shutter blade movement to close from a 90% exposure ^a to 10% exposure.	Transfer Close: the time for the shutter to move from 80% exposure ^a to 20% exposure.
MOP	Minimum Open Pulse: Minimum pulse width as driven with the SC30 or KSC101 controller.	Minimum Open Pulse: the minimum pulse width supplied by the SC30 controller.
MSOP	Minimum Shutter Open Pulse: Minimum shutter open time (pulse) using the MOP time from the SC30 or KSC101 controller.	Minimum Shutter Open Time: the minimum time the shutter will remain open when using the minimum open pulse (MOP) from the SC30 controller.

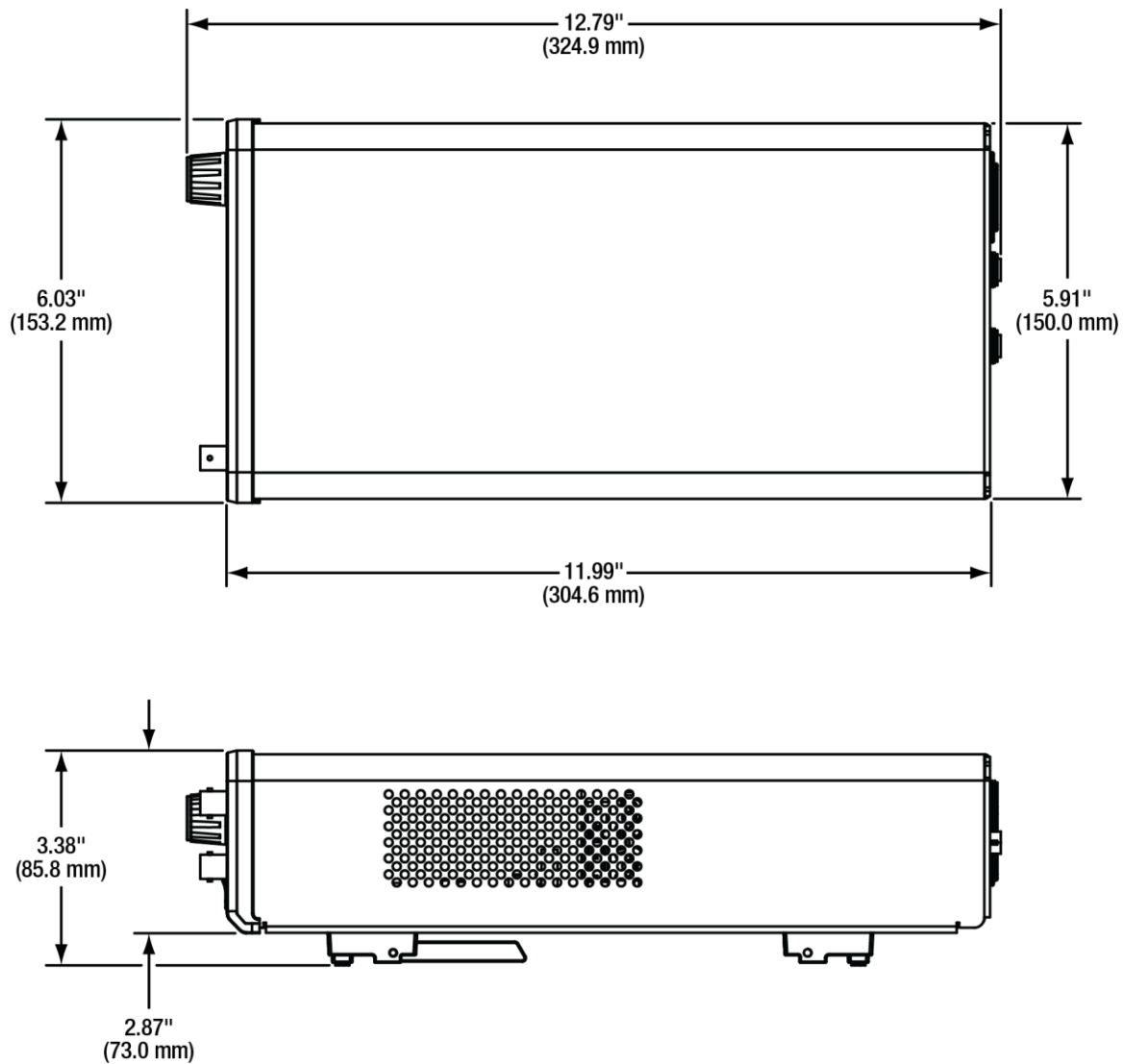
a. Exposure refers to the area of the shutter that is open.

Power Supply	
Supply Type	Switch Mode AC/DC
Input Voltage	100 - 240 VAC, 50 - 60 Hz
Power	60 VA
Fuse Rating	500 mA, 250 VAC
Fuse Type	Type T Slow Blow
Operating Temperature	10 to 40 °C
Storage Temperature	0 to 85 °C

Physical Features	
Dimensions	12.79" x 6.03" x 3.38"
Weight	3.35 lbs ^a
Input and Output Connectors	BNC
Display	3.5" TFT LCD Touchscreen
Adjustment Control	Encoder with RGB Backlight
USB Port Type	USB 2.0 Type-B

a. Just the Controller Console

1.4.2 Mechanical Drawing



1.5 Components

1.5.1 Included Components

- SC30 Shutter Controller
- Operating Manual
- 120 VAC US Power Supply Line Cord, when purchased in the US or 230 VAC Power Supply Line Cord for Europe
- 500 mA Fuse (preinstalled)
- USB 2.0 Type A to Type B Cable

1.6 Declaration of Conformity


THORLABS
www.thorlabs.com

EU Declaration of Conformity

Manufacturer: Thorlabs Inc.

Address: 43 Sparta Ave, Newton, 07860, USA

We hereby declare under our sole responsibility that:

Product: SC 30

Product description: Dual Channel Shutter Controller

is/are in conformity with the following directive(s):

2014/30/EU	Electromagnetic Compatibility (EMC) Directive
2014/35/EU	Low Voltage Directive (LVD)
2011/65/EU	Restriction of Use of Certain Hazardous Substances (RoHS), including 2015/863

and (harmonized) standards / technical specifications :

EN 61326-1	Electrical Equipment for Measurement, Control and Laboratory Use - EMC Requirements	2013
EN 61010-1	Safety Requirements for Electrical Equipment for Measurement, Control and Laboratory Use	2010 + A1:2019 + AC:2019
EN IEC 63000	Technical documentation for the assessment of electrical and electronic products with respect to the restriction of hazardous substances	2018

I hereby declare that the equipment named has been designed to comply with the relevant sections of the above referenced specifications and complies with all applicable Essential Requirements of the Directives.

Signed: 

Name: Danielle Strong **On:** 1/23/2025

Position: Director of Compliance



1.7 FCC Statement

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

Chapter 2 Safety

**Warning: Risk of Electrical Shock**

High voltage inside. To avoid electrical shock, before powering the unit, make sure that the protective conductor of the 3-conductor power cord is correctly connected to the protective earth contact of the socket outlet. Improper grounding can cause electrical shock resulting in severe injury or even death.

**Caution**

This unit must not be operated in explosive environments. Do not operate in wet or damp conditions. Do not obstruct the air-ventilation slots in the housing!

**Warning**

The unit is supplied with either a 115 V parallel blade line cord for North American use or a European compatible 220 V line cord. For all other applications use an IEC 320 compatible line cord fitted with a plug appropriate for your AC wall socket. Do not replace detachable mains lead with an inferior cord. This may cause a failure, damage, shock, or injury. Changing of the mains fuse can be done by the user, refer to Section 6.2 on page 30 for proper fuse selection. With the exception of the mains fuse and the connector (for cleaning), there are no user serviceable parts in this product.

NOTICE

Mobile telephones, cellular phones or other radio transmitters should not be used within the range of three meters of the unit since the electromagnetic field intensity may exceed the maximum allowed disturbance values according to EN50082-1.

NOTICE

The maximum USB and BNC cable length is 3 meters to avoid a susceptibility to RF interference according to IEC61000-4-3.

NOTICE**Note**

If equipment is used in a manner not specified by Thorlabs, the protection provided by the equipment may be impaired.

NOTICE**ATTENTION**

This equipment has been tested and found to comply with all requirements of the Canadian Interference-Causing Equipment Standard ICES-003 for digital apparatus.

NOTICE**ATTENTION**

Thorlabs GmbH is not responsible for any radio television interference caused by modifications of this equipment or the substitution or attachment of connecting cables and equipment other than those specified by Thorlabs GmbH. The correction of interference caused by such unauthorized modification, substitution, or attachment will be the responsibility of the user.

This device can only be returned when packed into the complete original packaging, including all packing inserts. If necessary, ask for a replacement package.

Chapter 3 Chassis Overview

Below is a short description of the outside of the SC30 Shutter Controller.

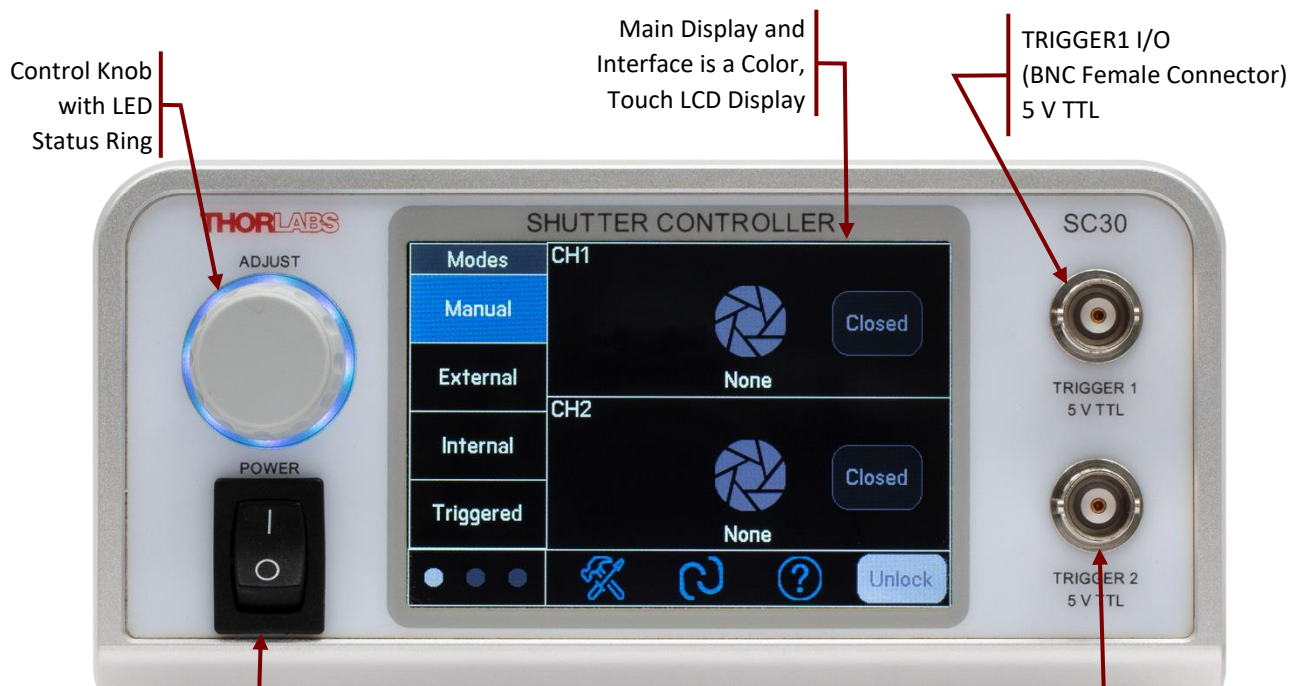


Figure 1 Front Panel

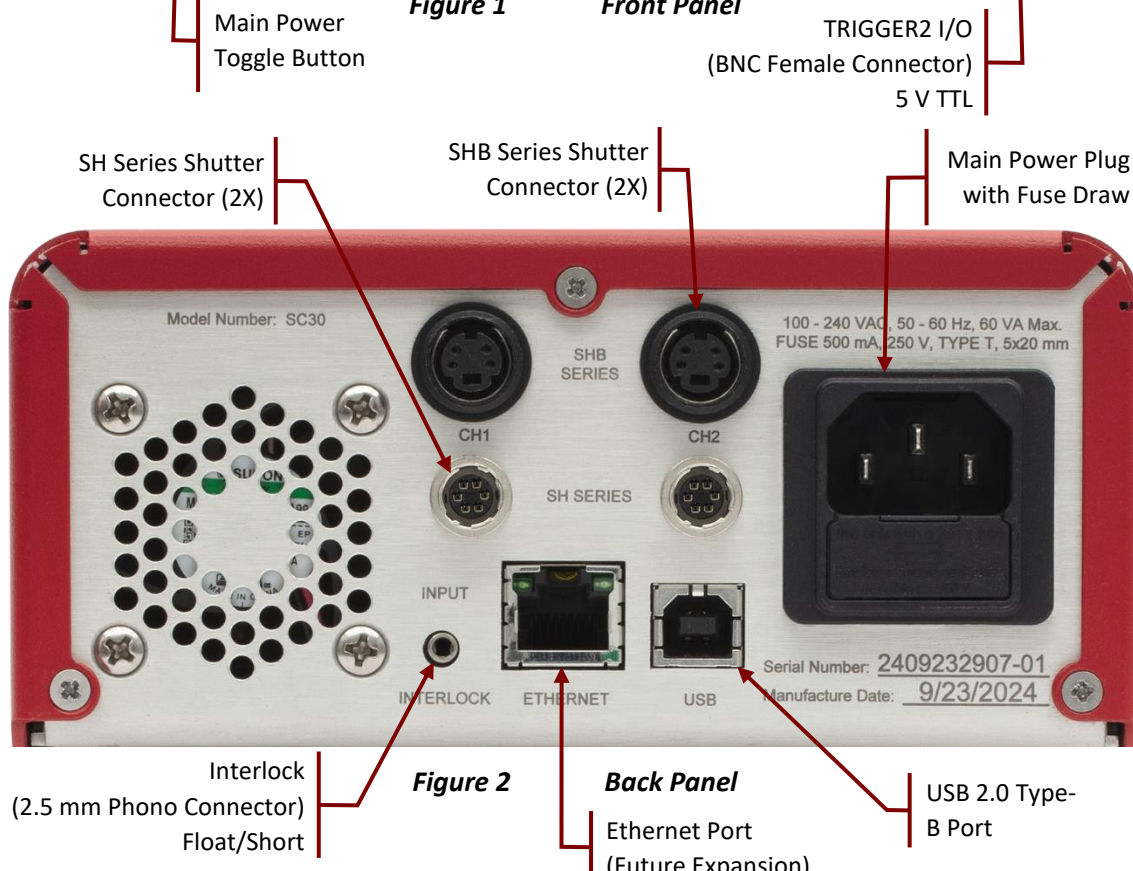


Figure 2 Back Panel

Chapter 4 Installation

The SC30 Shutter Controller is designed to operate from 85 to 264 VAC, and there is no need to make any adjustments as the switching power supply will determine the correct voltage automatically. Use the proper line cord for your region.

1. Locate the unit on a dry, level working surface. Verify that the ventilation vents are not obstructed.
2. Make sure the POWER switch on the front of the unit is in the OFF position.
3. Plug the female end of the AC line cord provided into the AC Input Receptacle on the rear of the unit. Plug the male end into a properly grounded AC socket.
4. Connect an appropriate shutter to channel 1 of the controller. If a 2nd shutter is available, connect to channel 2. Both devices will be supplied with their own cables. The connections can be found on the rear panel labelled “CH1” and “CH2”. While the device allows for 4 devices (2 x SH series and 2 x SHB series), only one can be selected for each channel through the touchscreen interface.
5. If an external signal is needed for operation in “EXTERNAL” or “TRIGGERED” modes, connect to “TRIGGER1” on the front panel. Note, the applied signal should be powered off when the SC30 power is off. The protection diodes allow for the signals to bleed between TRIGGER1 and TRIGGER2.
6. Turn the power switch on. The LCD will display a THORLABS splash screen before switching to the main operation screen.
7. The LCD is a touchscreen. Simply select the parameter you wish to adjust (Device Type, Mode of Operation, Trigger BNC IO, and Trigger Sources) and adjust using the displayed popup keypad or the control knob. Note that the control knob will be color coded to the function.

Chapter 5 Operation

5.1 Touchscreen LCD Display and Control Knob

The SC30 controller uses an easy-to-use touchscreen interface and a rotary knob with an LED indicator ring to make system adjustments. Any one of the four parameters can be selected by single tapping the display over the parameter. The control knob will modify the value by the increment set in the Settings menu (see Section 5.1.2). A double tap on the parameter will activate the keypad popup.

5.1.1 Home Screen

Figure 3 below shows the basic home screen when the SC30 controller is in a normal operating state. Each of the selections on the left side can be adjusted simply by tapping the desired. The 3 dots on the bottom left are used to select the left side menu options. The first group is the default state and will show the mode selection menu. The middle dot will bring up the save profiles settings and the final will show the system options.

The mode toolbar is located along the bottom. There are 3 selections. The first, , enters the setting and setup for the currently selected mode. This will allow the user to setup timing, triggers, sync operations and BNC allocations for the currently selected mode. Settings will be remembered when switching modes. Next,  toggles the Sync mode on and off. On shows linked chains while off shows broken chains. The sync setting sets how CH2 works with respect to CH1. Lastly,  provides user help.

The right section provides a graphical representation of the Shutter operations.

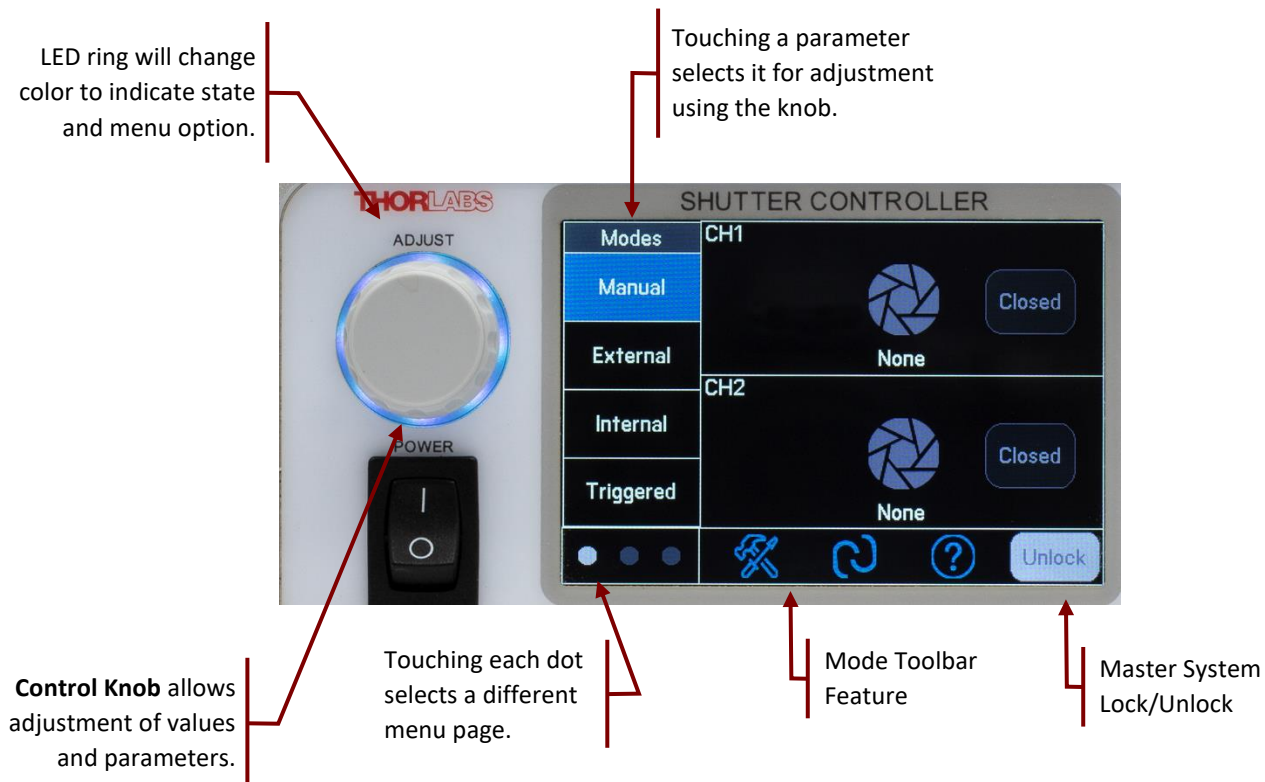


Figure 3 Home Screen – Default Manual Mode

5.1.2 Mode Settings Screen

When selecting the mode settings icon, a new menu screen will appear allowing for direct adjustment of the setting value for the currently selected mode; see Figure 4 below. In the example below, touching the down arrow will expand the list of choices. Some settings use number boxes. These can be adjusted by a single touch followed by knob adjustment, or a double tap to bring up the number pad popup screen.

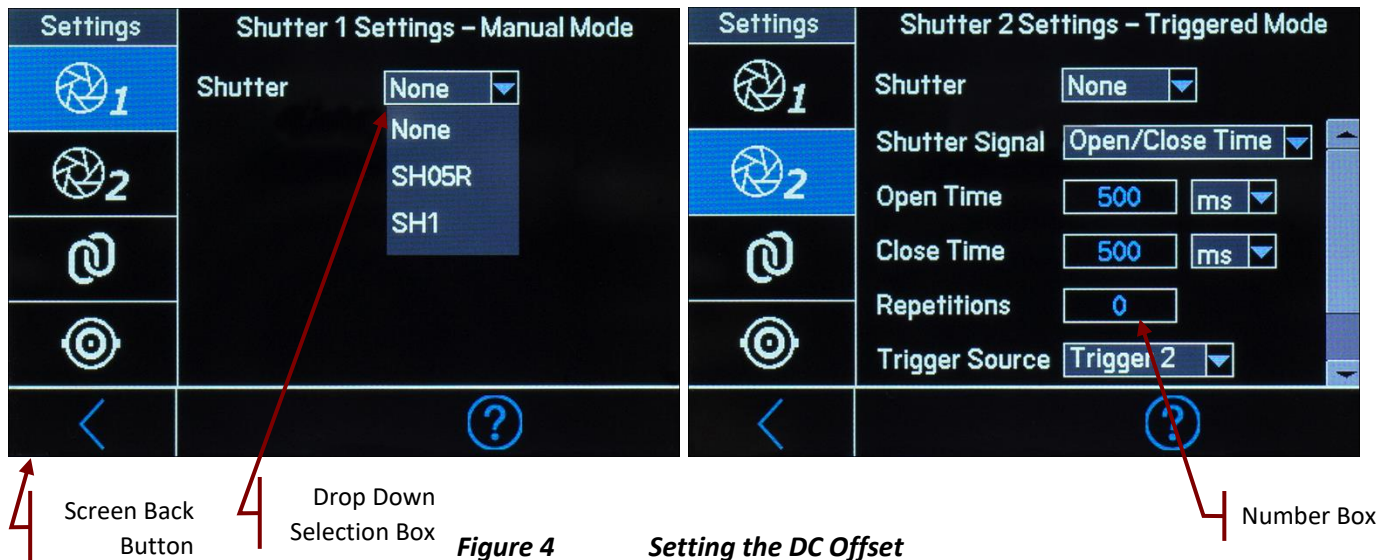


Figure 4 *Setting the DC Offset*

The other menus and their functions are:

-  and : These menu options allow the user to set the connected device, timing parameters, Repetition number, and trigger settings. Note only settings relevant to the selected mode will be shown at any given time.
- : This configures the Sync settings for the current mode. Sync allows the user to link CH2 to CH1. The user can set CH2 to be equal, inverted or delayed from CH1.
- : This configures the TRIGGER1 and TRIGGER2 BNC's. Either can be set as inputs or outputs. Outputs can be set to a specific output such as CHx drive signal or CHx monitor signal.
- **Back Icon:** Return to the previous screen.
- **Drop Down Box:** Provides a selection of choices. Touch the Down Arrow to the right to expand the selections. Touch an item to make the selection.
- **Number Box:** The number box is used to number selections such as open and close times, delays, etc. Touch the box to make the focus. Either use the knob to adjust values or double click to bring up a keypad popup to manually enter values.

Generally, the same adjustment screen will be available for each parameter chosen and adjusting their value is the same. The color of the selection knob will change to match the parameter selected. These colors can be changed in the Settings Menu, see page 18.

Some menu options such as “Lock/Unlock” and “Open/Close” are toggle buttons. There is no knob color or submenu. They will toggle between states when pressed.

5.1.3 Description of Parameters

Manual Mode

Manual mode can be selected on the menu of the home screen. It is the simplest operation available. There is no timing or triggering involved. Shutters are activated or closed by the user simply pressing the open/closed button for the desired channel. The sync function may be used to connect the operation of the 2 channels together. PC communications may also be used to control the states of the shutters.

External Mode

External mode is similar to manual mode; however, it relies on an externally applied TTL signal to operate the shutters. CH1 will always be considered the dominant channel and will force the TRIGGER1 BNC to be the driving source. If sync is used to control CH2 or if CH2 is unpopulated, the TRIGGER2 BNC can be set as an output. If CH2 is set to independent operation, it would use TRIGGER2 as its driving source. For operation, applying a high voltage to the BNC will open the corresponding shutter, while a low will close the device. This mode is equivalent to the SC10 X-GATE mode.

Internal Mode

Internal mode allows the user to free run the shutter without the need for a trigger source. The controller uses the open and close time values to control the operation of the shutter and generate an operating frequency. In this case, both TRIGGER BNC's default to inactive and can be set to generate output triggering to other devices. Operation is started and stopped using the system Lock/Unlock control. This is equivalent to the SC10's Auto Internal mode of operation.

Triggered Mode

Triggered mode utilizes the open and close times, triggering and repetition to generate an output sequence whenever a trigger is applied. In this case, the rising edge generates an event. The trigger source can be a TTL rising edge on TRIGGER1, a PC generated command, or even a front panel touch from the user.

Open and Close Time

Open time is the time the shutter installed will remain open while close time will set the close time for the connected shutter. This is used in both the Internal and Triggered modes to set the timing parameters for the shutter operation. This is the default timing operating condition; however, this can be changed to use frequency and duty cycle instead. To do this go to the shutter settings using the  symbol and change the "Shutter Signal" setting to toggle to the appropriate preference.

Unlock / Lock

The Unlock/Lock button located on the bottom right of the touchscreen is used as the system arming. It will enable or disable all shutter operation, regardless of mode, state of shutters or PC connection commands. When deactivated, all shutters will be forced to the closed state blocking all light sources. This can be used as an emergency stop or a system safe setting. If the interlock feature is enabled in the settings, the rear panel interlock will act as part of this function.

Repetition

Repetition is used primarily in the triggered mode to set the number of times a sequence will repeat whenever a trigger event occurs. By default, this is set to 0 indicating the repeat operation is disabled. Setting this to any number will enable the function and set the number of iterations.

Sync

The Sync functionality determines how CH2 works together with CH1. By default, the Sync setting is configured for independent operations. Touch the  icon in the mode toolbar along the bottom of the screen to activate sync. To adjust sync settings, touch the  and select the sync function  on the left side menu. A few options are available:

- Equal – If chosen, CH2 will track the same value as CH1.
- Inverted – CH2 will track CH1 but the state will be opposite. When CH1 is open, CH2 will be closed
- Delayed – CH2 will follow CH1 but there will be a time delay before state changes.

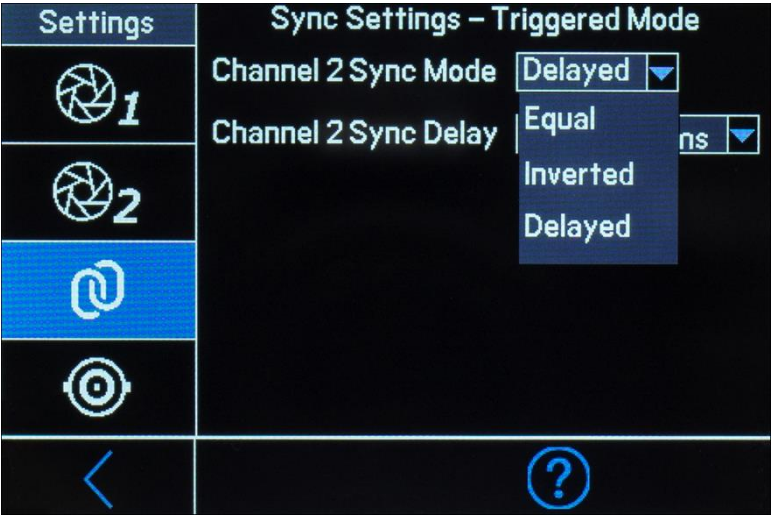


Figure 5 Sync Settings Screen

5.2 Profiles Menu

The Profiles menu allows the user to save their current settings as the default (Save menu option) or as a new profile (Save profile menu option) for easy recall later. This allows multiple devices to be set up and used with the SC30 controller and their settings recalled quickly. Profile options can be found by selecting the center dot on the bottom left corner of the screen. See Figure 6 below.



Figure 6 Profiles Menu Screen

The Operations dropdown menu has five options, and their functions are:

- **Startup Profile:** Saves all the settings, except knob color, to be loaded every time the device starts up.
- **Save Profile:** Saves the settings to one of five profiles, which can be recalled as needed.
- **Load:** Loads the settings from one of the saved profiles. There is a permanent profile to restore factory defaults located at the bottom of the list.
- **Delete:** Deletes the selected profile

The SC30 controller can save up to five profiles. To save a profile, tap the Profiles menu option, tap the Save Profile option, and select a profile to change. This will bring up a keyboard window to enter a new profile name. Select OK when done to save your profile. If the profile number selected has already had a profile saved, it will be replaced.

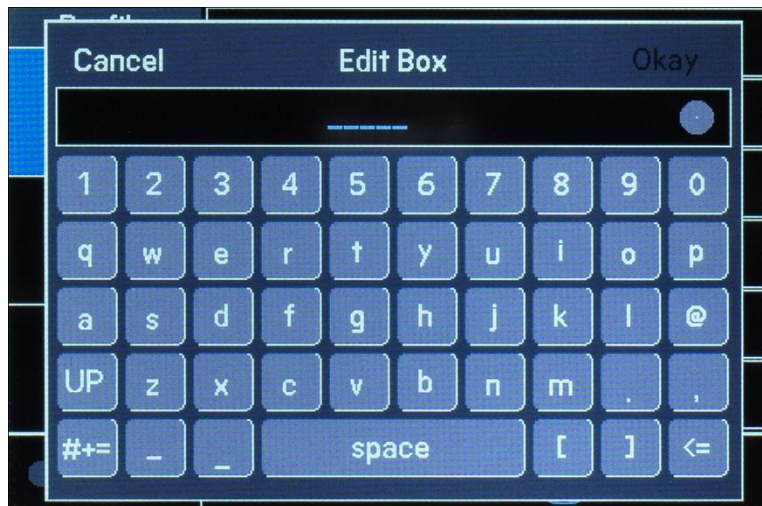


Figure 7 **Settings Keyboard**

5.3 Options Menu

Pressing the third dot on the bottom left corner of the main screen will bring up the Options menu. The options - menu populates the side menu categories with system wide settings. For the SC30, there is only one settings page.

The first option, Display, lists system settings directly related to how information is displayed. Knob allows the user to set up the control knob operations. The General tab allows the user to change miscellaneous system settings. Lastly, About provides system information such as serial numbers and firmware versions.

5.3.1 Display Sub-Menu

The Display sub-menu allows for adjustment of user preferences as listed below.

- **Double Tap Timeout:** This determines how sensitive double tapping the display responds. By default, 250 ms sets a balanced approach. Lower this number if you prefer a faster double tap. Raise this if you are finding double tap does not activate.
- **Disable Channel Parameters:** This option allows the user to disable the parameter text shown on the display when in the internal or triggered modes. Checking the box will disable this text.

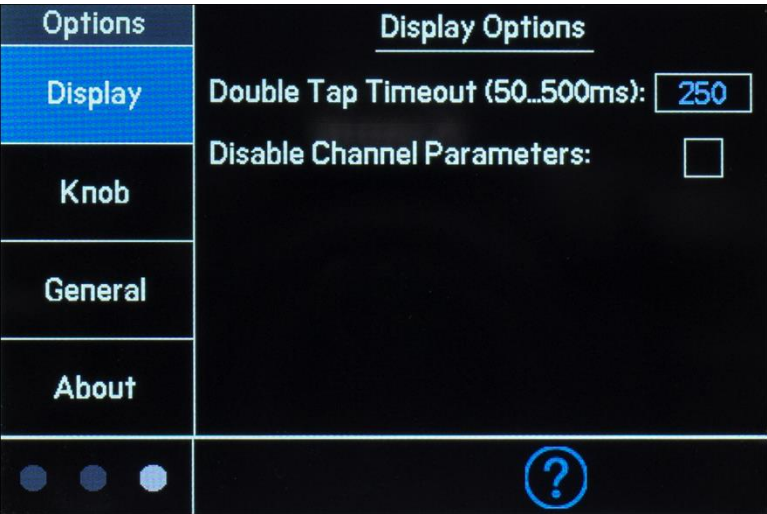


Figure 8 Display Options Menu

5.3.2 Knob Menu

The Limits menu allows the user to set minimum and maximum limits on DC Offset, Trigger location, Rise Time, and Amplitude. This allows safety limits to be enforced preventing cells from being overdriven.

- **Disable Knob LED's:** Checking the box will disable the Knob LED. Default state is unchecked allowing LED operation.
- **Knob Brightness:** Sets the brightness of the knob LED's between 10 and 100%. Default is 80%.
- **Enable Knob Dimming:** This function will cause the LED brightness to drop to a predefined level when a period of inactivity occurs. It is useful for environments that want to minimize light as much as possible. Defaults top disabled or unchecked.
- **Knob Dimming Timeout:** This sets the time or inactivity to activate the dimming. By default, this is set to 30 seconds. This feature is hidden until Enable knob dimming is checked.
- **Knob Dim Brightness:** Sets the dimmed brightness. Defaults to 50%. This feature is hidden until Enable knob dimming is checked.

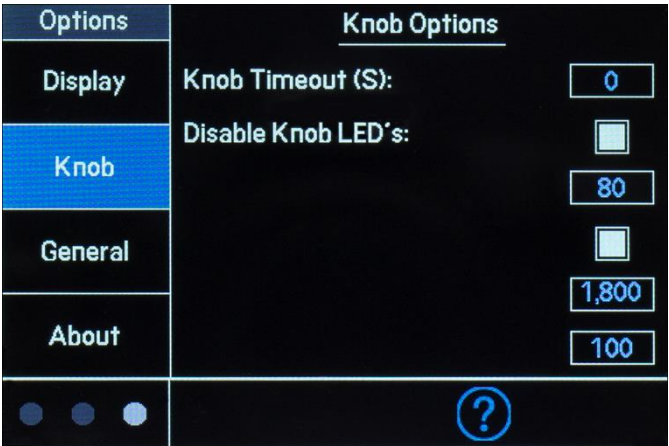


Figure 9 Knob Options Menu

5.3.3 General Menu

The General Controls menu is primarily used to setup user preferences.

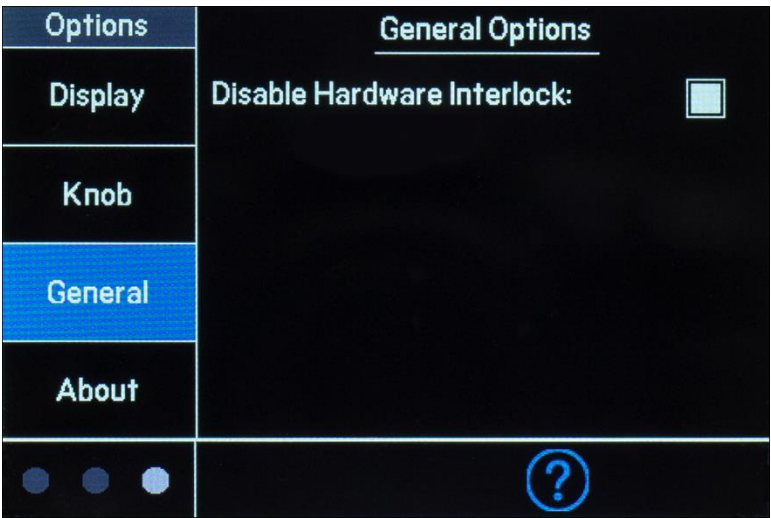


Figure 10 General Options Menu

- **Disable Hardware Interlock:** This checkbox enables the use of the interlock input, located on the rear panel. When active, the shorting the interlock will disable the system as if using the Unlock button and provide a popup alert.

5.3.4 About Tab

The About Tab is an information tab that displays the company name, part name, serial number, and the firmware revisions. There is no interactable content on this page.

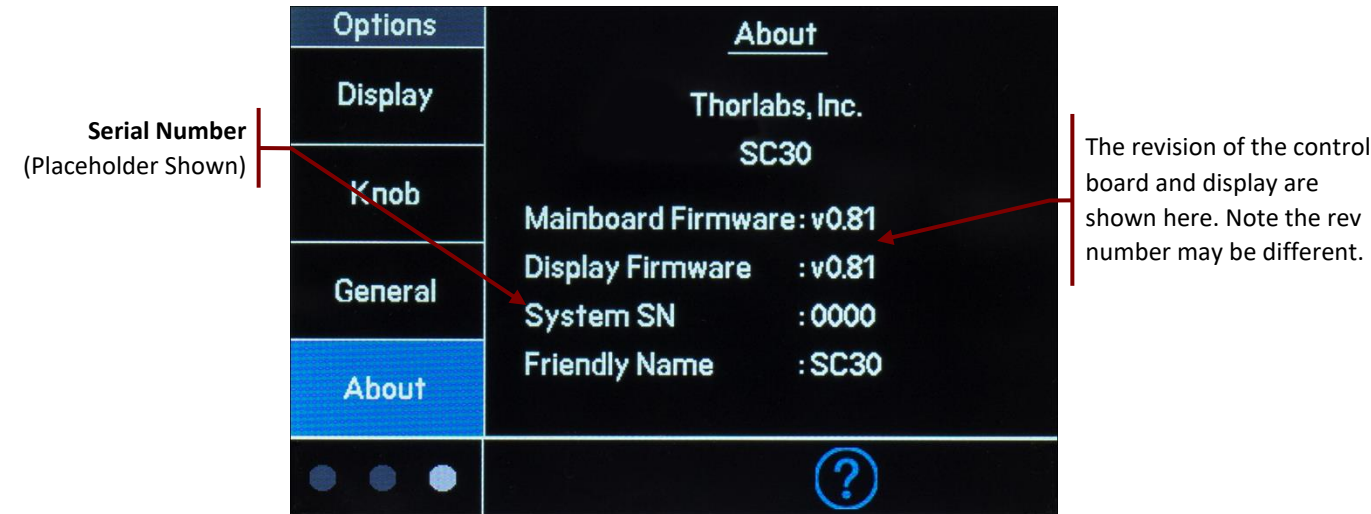


Figure 11 About Tab

5.4 Help Icon

The user can get help on any of the settings menu options by tapping the question mark icon at the bottom right of the screen. See Figure 8 on page 18. Note that this help is only available on the various Settings tabs wherever the question mark icon is shown. After tapping the icon, it will change to a question mark icon with an arrow.

While active, the help mode will disable selection of any of the options. Instead, tapping any menu, value, or selection option button on the screen will bring up a help popup window. The user needs to tap something to exit out of help mode, even the help icon. For example, tapping the question mark icon then tapping the Signals Tab will bring up the help popup screen shown in Figure 12 below.

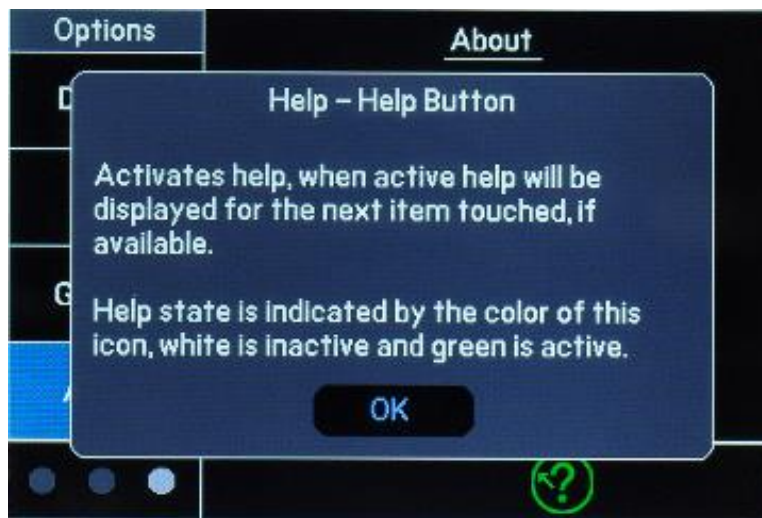


Figure 12 **Signals Help Popup Window Example**

To close the popup window and exit the help mode, tap on the OK on-screen button.

5.5 Front Panel BNC Connections

5.5.1 TRIGGER1 BNC

This connection can be either an input or an output based on the mode of operation and user preferences. In general, TRIGGER1 is locked to the trigger input function for CH1. It acts as the dominant input for the external and triggered modes of operation. When not actively forced such as the manual and internal modes, this can be set as an output that can be configured for either driving signal or shutter monitor signal. The BNC signal is buffered through a 100 Ω resistor with protection diodes between GND and the 5 V supply rails. Normal operation is 5 V TTL. Above and below this range, the resistance will current limit and the diodes will shunt signal to the rails. It should be noted that applying a signal to these inputs when the device is unpowered will result in signal bleed between TRIGGER1 and TRIGGER2 due to the protection diodes. This will not damage the device.

5.5.2 TRIGGER2 BNC

Trigger2 operates much like Trigger1. It is only forced into a state when using external when sync is disabled. By default, it will be set up as an output set to Ch1 drive signal.

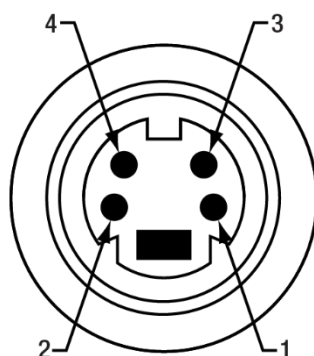
5.6 Rear Panel BNC Connections

5.6.1 CH1 and CH2 DIN Connectors

The interface connector to the SHB series on the SC30 is a 4-pin mini-DIN shielded female connector that accepts the 4-pin mini-DIN male connector.

The pin descriptions are as follows:

Pin	Description
1	Shutter Detect
2	Modulation +
3	+5 VDC
4	Modulation -
SHIELD	GND

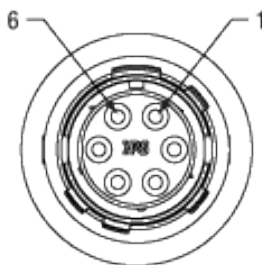


SC30 Mini-DIN Connector Pin-Out

5.6.2 CH1 and CH2 Hirose Connectors

The interface to the SH05 and SH1 shutters is via a 6-pin HIROSE female connector. The pin descriptions are as follows:

Pin	Description
1	Opto - Anode (12 V limited to 20 mA)
2	Shutter Coil - 24 V pulse - 10 V steady state (0.4 A max)
3	Shutter Coil - GND (ON) - Open Ckt (OFF)
4	Opto - Cathode -GND
5	Opto - Emitter- GND
6	Opto - Collector (2.5 V)



SC30 Hirose Connector Pin-Out

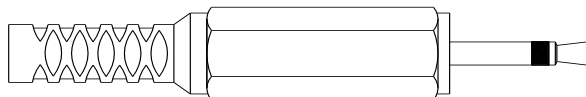
5.6.3 Interlock Connector

The SC30 is equipped with a remote interlock connector located on the rear panel (Refer to Figure 2). In order to enable the controller, a short circuit must be applied across the terminals of the remote interlock connector. In practice this connection is made available to allow the user to connect a remote actuated switch to the connector (i.e. an open-door indicator). The switch (which must be normally open) must be closed in order for the unit to be enabled. Once the switch is in an open state and a shutter is connected, the shutter will automatically close, and the display will flash “INTERLOCK OPEN”. If the switch returns to a closed condition the shutter must be re-enabled at the unit by pressing the ENABLE switch.

All units shipped from Thorlabs are configured with a shorting pin installed in the Interlock connector. If you are not going to use this feature, then you can leave the shorting pin installed and the unit will operate normally as described in the procedures above. This feature can be enabled or disabled in the Options → General menu. By default, this feature is disabled.

If you wish to make use of the Interlock feature you will need to acquire the appropriate connector and wire it to your remote interlock switch. Next, remove the shorting device from the back of the SC30 and install the connector. Make sure to activate the feature in the Options → General menu.

The Interlock input only accepts a 2.5 mm mono phono jack. This connector is readily available at most electronics stores.



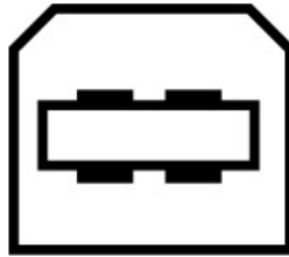
Remote Interlock Connector

The electrical specifications for the Interlock Input are as follows:

Specification of Interlock Input	
Type of Mating Connector	2.5 mm Mono Phono Jack
Open Circuit Voltage	+5 VDC with Respect to Chassis Ground
Short Circuit Current	0.5 m ADC
Connector Polarity	+ 5 V, Barrel is Ground
Interlock Switch Requirements	Must be NO Dry Contacts Under No Circumstances Should Any External Voltages be Applied to the Interlock Input.

5.6.4 USB Connector

The SC30 utilizes a USB type-B connector to allow remote control of the device while used with a computer. See Section 5.7 for operation through the USB interface and the full commands list. A type-A to type-B USB cable is provided with the controller.



USB 2.0 Type-B Port

5.6.5 Ethernet Connector

An ethernet connection is supplied on the rear panel, however it is not currently operational. This is planned for a future upgrade and will eventually be available through a firmware update.

5.7 Computer Control Operation

5.7.1 GUI Operation

A Windows GUI application software is provided to the user to control the Shutter Controller from the computer. A Software Development Kit (SDK) also comes with the device, which will allow the user to integrate the Shutter Controller into his own applications.

5.7.2 Software Requirements

- Windows 7, 64-bit or later
- .Net Framework 4.5 (or higher) must be installed
- Microsoft Visual Studio for SDK Demo project

5.7.3 Installation

The installation package is available from the SC30 Shutter Controller product page on the Thorlabs, Inc. web site for users to download. The installation includes two components: Shutter Control Software and Thorlabs Device SDK, as shown in Figure 13.



Figure 13 Shutter Controller Software Installation

For the first time installation, both components are checked and must be installed. The setup checks if another version of the selected component was installed. If another version exists, the new installation cannot complete until another version is removed.

The following files are installed in the following directories:

1. GUI application executable and dependencies (DLLs) are installed in
C:\Program Files (x86)\Thorlabs\TL_Devices\SC30
2. Thorlabs Device Plugin is stored in folder
C:\Program Files (x86)\Thorlabs\TL_Devices\TL_Device_Plugins
3. Thorlabs Device SDK files are stored in folder
C:\Program Files (x86)\Thorlabs\TL_Devices\TL_Device_SDK

Thorlabs Device Dynamic-Link Library (TLDeviceSDK.dll) and its dependency are required by the GUI application, or any other applications developed with the Thorlabs Device SDK. To make it convenient to run the executables without copying the DLLs, the full path to TLDeviceSDK.dll, C:\Program Files (x86)\Thorlabs\TL_Devices\TL_DEVICE_SDK\Bin is added to the Windows **PATH** environment variable.

5.7.4 Command Line Operation

The system may also be controlled by a command line language through the USB port. This is offered to enable operation through a terminal interface or for those who may want to write their own program to control the unit. The command language is described below. Prior to running the command line interface, the unit should be powered on, and a USB cable should be connected between the SC30 controller and the host.

The terminal emulator should be set as follows:

Terminal Emulator Settings	
Baud Rate	115.2 kb/s
Data Bits	8
Parity	None
Stop Bits	1
Flow Control	None

If you are using a terminal program, be sure to check the option “Append line feeds to incoming line ends”, or a similar option, depending upon the program you are using. In HyperTerminal you can find this option by navigating to File → Properties, selecting the Settings Tab and clicking “ASCII Setup...” button. It is the first option in the “ASCII Receiving” box.

If the connection is correct, you will see the following after pressing the Enter key.

5.7.5 Keywords (Commands and Queries)

The following table describes commands and queries:

Command Syntax	Description
?	Returns a list of all available commands
Standard Menu (?, 1)	
ID?	Returns product header and firmware version
model?	Get device model
serial?	Returns device serial number
fwver?	Get firmware version
friendly?	Get Device Friendly Name
friendly=n	Set Device Friendly Name (alphanumeric)
unlocked?	Get unlocked state
unlocked=n	Set unlocked state
mode?	Get active mode
mode=n	Set active mode (0 = Manual, 1 = External, 2 = Internal, 3 = Triggered)
syncenable?	Get sync enabled state
syncenable=n	Set sync enabled state (0 = disabled, 1 = enabled)
mansyncmode?	Get manual sync mode
mansyncmode=n	Get manual sync mode (0 = equal, 1 = inverted, 2=delayed)
mansyncdelay?	Get manual sync delay in ms/s
mansyncdelay=n	Set manual sync delay (n=Delay in ms/s)
mansyncdelayuom?	Get manual sync delay units of measurement (uom) in ms/s
mansyncdelayuom=n	Set manual sync delay units of measurement (0=ms, 1=s)

Standard Menu (?,2)	
extsyncmode?	Get external sync mode
extsyncmode=n	Get external sync mode (0 = equal, 1 = inverted, 2=delayed)
extsyncdelay?	Get external sync delay in ms/s
extsyncdelay=n	Set external sync delay (n=Delay in ms/s)
extsyncdelayuom?	Get external sync delay units of measurement (uom) in ms/s
extsyncdelayuom=n	Set external sync delay units of measurement (0=ms, 1=s)
intsyncmode?	Get internal sync mode
intsyncmode=n	Get internal sync mode (0 = equal, 1 = inverted, 2=delayed)
intsyncdelay?	Get internal sync delay in ms/s
intsyncdelay=n	Set internal sync delay (n=Delay in ms/s)
intsyncdelayuom?	Get internal sync delay units of measurement (uom) in ms/s
intsyncdelayuom=n	Set internal sync delay units of measurement (0=ms, 1=s)
trigsyncmode?	Get triggered sync mode
trigsyncmode=n	Get triggered sync mode (0 = equal, 1 = inverted, 2=delayed)
trigsyncdelay?	Get triggered sync delay in ms/s
trigsyncdelay=n	Set triggered sync delay (n=Delay in ms/s)
trigsyncdelayuom?	Get triggered sync delay units of measurement (uom) in ms/s
trigsyncdelayuom=n	Set triggered sync delay units of measurement (0=ms, 1=s)
enablex?	Get Enable channel x, where x = 1 or 2
Standard Menu (?,3)	
enablex=n	Set Enable channel x, where x = 1 or 2 (0=disabled, 1=enabled)
triggerstatex?	Get the trigger state for channel x, where x = 1 or 2
swtrigx?	Get Software trigger for channel x, where x = 1 or 2
swtrigx=n	Set Software trigger for channel x, where x = 1 or 2 (0=Inactive, 1=active)
shutterstatex?	Get shutter state for channel x, where x = 1 or 2. (0=closed, 1=open)
intsignalx?	Get internal signal for channel x, where x = 1 or 2.
intsignalx=n	Set internal signal for channel x, where x = 1 or 2. (n = 0 to 1)
intfreqx?	Get internal frequency for channel x, where x = 1 or 2.
intfreqx=n	Set internal frequency for channel x, where x = 1 or 2. (n = 0.1 to 50.0)
intdutyx?	Get internal duty cycle for channel x, where x = 1 or 2.
intdutyx=n	Get internal duty cycle for channel x, where x = 1 or 2. (n = 0.0 to 100.0)
intopenx?	Get internal open time (ms/s) for channel x, where x = 1 or 2.
intopenx=n	Set internal open time (ms/s) for channel x, where x = 1 or 2. (n=Time)
intopenuomx?	Get internal open time unit of measurement for channel x, where x = 1 or 2.

intopenuomx=n	Set internal open time unit of measurement for channel x, where x = 1 or 2. (0=ms, 1=s)
intclosex?	Get internal close time (ms/s) for channel x, where x = 1 or 2.
intclosex=n	Set internal close time (ms/s) for channel x, where x = 1 or 2. (n=Time)
intcloseuomx?	Get internal close time unit of measurement for channel x, where x = 1 or 2.
intcloseuomx=n	Set internal close time unit of measurement for channel x, where x = 1 or 2. (0=ms, 1=s)
Standard Menu (?,4)	
intcyclecntx?	Get internal cycle count for channel x, where x = 1 or 2.
intcyclecntx=n	Set internal cycle count for channel x, where x = 1 or 2. (n=Cycle Count)
trigsignalx?	Get triggered signal for channel x, where x = 1 or 2.
trigsignalx=n	Set triggered signal for channel x, where x = 1 or 2. (0=open/close, 1=freq/duty)
trigfreqx?	Get triggered frequency for channel x, where x = 1 or 2.
trigfreqx=n	Set triggered frequency for channel x, where x = 1 or 2. (n=0.1 to 50.0)
trigdutyx?	Get triggered duty cycle for channel x, where x = 1 or 2.
trigdutyx=n	Set triggered duty cycle for channel x, where x = 1 or 2. (n=0.00 to 100)
trigopenx?	Get triggered open time (ms/s) for channel x, where x = 1 or 2.
trigopenx=n	Set triggered open time (ms/s) for channel x, where x = 1 or 2. (n=Open Time)
trigopenuomx?	Get triggered open time unit of measurement for channel x, where x = 1 or 2.
trigopenuomx=n	Set triggered open time unit of measurement for channel x, where x = 1 or 2. (0=ms, 1=s)
trigclosex?	Get triggered close time (ms/s) for channel x, where x = 1 or 2.
trigclosex=n	Set triggered close time (ms/s) for channel x, where x = 1 or 2. (n=Open Time)
trigcloseuomx?	Get triggered close time unit of measurement for channel x, where x = 1 or 2.
trigcloseuomx=n	Set triggered close time unit of measurement for channel x, where x = 1 or 2. (0=ms, 1=s)
trigcyclecntx?	Get triggered cycle count for channel x, where x = 1 or 2.
trigcyclecntx=n	Set triggered cycle count for channel x, where x = 1 or 2. (n=Cycle Count)
trigtrigsrcx?	Get triggered trigger source for channel x, where x = 1 or 2.
Standard Menu (?,5)	
trigtrigsrcx=n	Get triggered trigger source for channel x, where x = 1 or 2. (0=TRIGGER1, 1=TRIGGER2, 2=Software)
trigtrigdlyx?	Get triggered trigger delay for channel x, where x = 1 or 2

trigtrigdlyx=n	Set triggered trigger delay for channel x, where x = 1 or 2 (n=Delay ms/s)
trigtrigdlyuomx?	Get triggered trigger delay unit of measurement for channel x, where x = 1 or 2
trigtrigdlyuomx=n	Set triggered trigger delay unit of measurement for channel x, where x = 1 or 2 (0=ms, 1=s)
shuttercyclecomp?	Get number of cycles completed for channel x, where x = 1 or 2
connected?	Get type of shutters detected for channel x, where x = 1 or 2 (Blade, Diaphragm)
shutterx?	Get selected shutter for channel x, where x = 1 or 2
shutterx=n	Set selected shutter for channel x, where x = 1 or 2 (n=Shutter Model)
shuttertypex?	Get active shutter type for channel x, where x = 1 or 2 (none, Blade, Diaphragm)
shutterinfox?	Get active shutter info for channel x, where x = 1 or 2 (All shutter parameters)
shuttererrorx?	Get shutter error for channel x, where x = 1 or 2 (all active errors)
interlock?	Get interlock state (0=inactive, 1=active)
interlockdisable?	Get interlock disable status
interlockdisable=n	Set interlock disable status (0=enabled, 1=disabled)
io_mantrig1?	Get IO Manual Mode Trigger 1 setting
io_mantrig1=n	Set IO Manual Mode Trigger 1 setting (0=inactive, 1=Input, 2=output+Ch1/2Driven, 3=output+Ch1 Driven, 4=output+Ch1 active, 5=output+Ch2 Driven, 6=output+Ch2 Actual)
io_mantrig2?	Get IO Manual Mode Trigger 2 setting
io_mantrig2=n	Set IO Manual Mode Trigger 2 setting (n=0, 1, 2, 3, 4, 5, 6)
Standard Menu (?,6)	
io_exttrig1?	Get IO External Mode Trigger 1 setting
io_exttrig2?	Get IO External Mode Trigger 2 setting
io_exttrig2=n	Set IO External Mode Trigger 2 setting (n=0, 1, 2, 3, 4, 5, 6)
io_intrig1?	Get IO Internal Mode Trigger 1 setting
io_intrig1=n	Set IO Internal Mode Trigger 1 setting (n=0, 2, 3, 4, 5, 6)
io_intrig2?	Get IO Internal Mode Trigger 2 setting
io_intrig2=n	Set IO Internal Mode Trigger 2 setting (n=0, 2, 3, 4, 5, 6)
io_trigtrig1?	Get IO Triggered Mode Trigger 1 setting
io_trigtrig1=n	Set IO Triggered Mode Trigger 1 setting (n=0, 1, 2, 3, 4, 5, 6)
io_trigtrig2?	Get IO Triggered Mode Trigger 2 setting
io_trigtrig2=n	Set IO Triggered Mode Trigger 2 setting (n=0, 1, 2, 3, 4, 5, 6)
dispparamdisable?	Get disable displaying parameters status
dispparamdisable=n	Set disable displaying parameters (0=enabled, 1=disabled)

load=n	Load Profile from Index (n=1..5)
save	Save Startup Settings
save=n, <Name>	Save Settings to Profile Index (n=1..5)
clear=n	Clear Profile from Index (1-5)
profnamex?	Get the name of the profile at index x
Standard Menu (?,6)	
profnamex=str	Set the name of the profile at index x as str text
knobledisable?	Get the Knob Backlight status (0=enabled, 1=disabled)
knobledisable=n	Set the Knob Backlight status (0=enabled, 1=disabled)
knobdimenable?	Get enable knob dimming status (0=disabled, 1=enabled)
knobdimenable=n	Set enable knob dimming status (0=disabled, 1=enabled)
knobctrltimeout?	Get the knob timeout in seconds
knobctrltimeout=n	Set the knob timeout in seconds (n=seconds)
knobdimtimeout?	Get the knob dimming timeout in seconds
knobdimtimeout=n	Set the knob dimming timeout in seconds (n=seconds)
knobbright?	Get the knob brightness percent
knobbright=n	Set the knob brightness percent (n = 0.0 to 100.0)
dbltaptimeout?	Get double tap timeout in milliseconds
dbltaptimeout=n	Set double tap timeout in milliseconds
activecfg?	Get the last loaded profile
restoredefaults=	Set Restore all settings to factory defaults

Chapter 6 Maintenance and Cleaning

6.1 Cleaning the SC30 Controller

There are no serviceable parts inside. Do not open.

The exterior of the unit can be cleaned with a soft, lint free damp cloth. If needed, a mild detergent can be used. No harsh chemicals or abrasives should be used.

6.2 Changing the Fuse

The SC30 controller uses a 500 mA, 250 VAC Littlefuse part number 0218.500 HXP fuse. Should the fuse blow and need to be replaced, the same fuse must be used. The fuse can be found in the tray below the AC mains plug.



Shock Warning

To avoid electrical shock and severe injury, before changing the fuse: turn the SC30 off, unplug the AC mains cord from the wall then the SC30. Failure to do so can cause electrical shock resulting in severe injury or even death.

To replace the fuse:

1. Remove the mains power cord.
2. Pull the fuse draw open.
3. Remove the old fuse and dispose appropriately.
4. Insert the new 500 mA Type T Slow Blow fuse into the tray.
5. Close the tray and plug in the mains power cord.

The unit can then be plugged back in and powered on. If the unit fails to start, refer to the Trouble Shooting section starting on page 31 or contact your local technical support office.

Chapter 7 Trouble Shooting

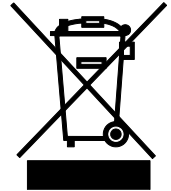
7.1 Fault Tracing

The following table describes some typical problems that may be encountered while using the SC30 controller and possible solutions to these problems.

Problem	Solution(s)
Unit does not turn on when switching the power switch to the ON position.	1. Make sure AC line cord is fully inserted into the AC Input receptacle and plugged into an outlet providing 100 to 240 VAC. 2. Fuse(s) may be open. Refer to Section 6.2 on page 30 for information on replacing open fuses. If the problem persists, please return the unit to Thorlabs for evaluation.
I can't connect to the SC30 controller over the USB com port.	1. Make sure that the com port is configured correctly for the unit. Refer to Section 5.7.3 on page 24 for the correct com port settings. 2. The incorrect com port is selected on your terminal program or SC30 application.

Chapter 8 Warranty and RMA Information

Thorlabs verifies our compliance with the WEEE (Waste Electrical and Electronic Equipment) directive of the European Community and the corresponding national laws. Accordingly, all end users in the EC may return “end of life” Annex I category electrical and electronic equipment sold after August 13, 2005 to Thorlabs, without incurring disposal charges. Eligible units are marked with the crossed out “wheelie bin” logo (see right), were sold to and are currently owned by a company or institute within the EC and are not disassembled or contaminated. Contact Thorlabs for more information. Waste treatment is your own responsibility. “End of life” units must be returned to Thorlabs or handed to a company specializing in waste recovery. Do not dispose of the unit in a litter bin or at a public waste disposal site. It is the user’s responsibility to delete all private data stored on the device prior to disposal.



Annex I

8.1 Return of Devices

This precision device is only serviceable if returned and properly packed into the complete original packaging including the complete shipment plus the cardboard insert that holds the enclosed devices. If necessary, ask for replacement packaging. Refer servicing to qualified personnel.

Chapter 9 Thorlabs Worldwide Contacts

For technical support or sales inquiries, please visit us at www.thorlabs.com/contact for our most up-to-date contact information.



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